

Generation Interconnection Request System Impact Study Report

For Queue Project Q14-0367

Bloomington 138 kV
July 26, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0367. The Interconnection Customer proposes a 13.4 MW generating facility interconnecting at the Bloomington 138 kV substation in the Virginia area. The request was received on August 24, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 170 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$169,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Taber - Glenannan 345 KV, the Adrian Energy Associates Llc 345-138 kV Autotransformer loads to 118.3% of its 519 MVA rating. Project DFAX: 5.4%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Adrian Energy Associates Llc 345-138 kV Autotransformer	Replace transformer with higher-rated unit	\$23,263,000	1% (\$169,000)
	Total allocated to project		\$169,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.